

Total No. of Questions : 10]

SEAT No. :

P3745

[5561]-142

[Total No. of Pages : 4

B.E. (Mechanical)

MACHANICAL SYSTEM DESIGN

(2012 Course) (402048) (Semester-II)

Time : 3 Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8, Q9 or Q10.
- 2) Neat diagrams must be drawn whenever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume suitable data, if necessary.

Q1) a) A machine tool requires six speed gear box. Which having a 160 rpm minimum and 1000 rpm maximum speed, when the motor shaft is 1440 rpm. Draw the speed diagram for the following structural formula.
i) 2 (3) 3 (1) ii) 2 (1) 3 (2). **[6]**

b) Give formula for calculating Mean and Standard deviation of resultant Population if two individual populations are to be added and multiplied. **[4]**

OR

Q2) a) Draw the structure diagram and identify the optimum structural formula out of them. i) 2 (1) 3 (2) ii) 2 (3) 3 (1) iii) 3 (2) 2 (1) **[6]**

b) Define design tolerance (DT) and natural tolerance (NT) in statistical design. State what happens when $NT=DT$, $NT>DT$ and $DT>NT$. **[4]**

Q3) a) Explain the concepts of Unit Load and Containerzation. **[4]**

b) A three idler, troughed belt, horizontal conveyor is to be used for transporting 500 ton of iron per hour having mass density of iron ore is 1700 kg/m³. If the belt speed is 2m/sec, determine the required belt width. Take surcharge factor = 0.1. **[6]**

OR

P.T.O.

Q4) a) State the appropriate guidelines for selection of material handling system. [4]

b) A flat horizontal belt conveyor is to be used for transporting material with mass density of 3 ton/m³. The belt is 750 mm wide and has speed of 2.75 m/s. Determine capacity of conveyor if surcharge angle is 25° ($k = 2.35 \times 10^{-4}$). [6]

Q5) a) Derive an expression for the thickness of a thick cylinder subjected to internal pressure by using maximum shear stress theory. [8]

b) A seamless cylinder with a storage capacity of 0.025 m³ is subjected to an internal pressure of 20 MPa. The length of the cylinder is twice its internal diameter. The cylinder is made of plain carbon steel 20C8 ($S_{ut} = 390 \text{ N/mm}^2$) and the factor of safety is 2.5. Determine the dimensions of the cylinder. [10]

OR

Q6) The following data refers to single acting hydraulic cylinder. [18]

Pressure of hydraulic fluid = 10 MPa

Operating force available at the piston rod. = 10 kN

Friction due to piston ring and stuffing box = 10% of operating force

Thickness of cylinder flange = 10 mm

Thickness of cylinder head = 8 mm

Cylinder and, cylinder head material = FG200

Modulus of elasticity for FG200 = 100 GPa

Thickness of Zinc gasket = 3 mm

Modulus of elasticity for zinc = 83 GPa

Number of bolts = 4

Preload in each bolt	= 2.8 KN
Bolt material	= FeE400
Modulus of elasticity for FeE400	= 207 GPa
Factor of safety for cylinder	= 5
Factor of safety for bolts	= 6
Standard diameter of cylinder	= 20, 30, 40, 50, 60 mm
Standard Thickness of cylinder	= 2, 4, 5, 6, 7, 8, 10 mm
Standard diameter of bolts	= 8, 10, 12, 14, 16, 18, 20, 22 mm

Determine :

- i) Inner diameter of cylinder
- ii) Thickness of cylinder
- iii) Diameter of bolts

- Q7)** a) What is the difference between centre and overhung crankshafts? [6]
- b) The bore of a cylinder of the four stroke diesel engine is 120 mm. The maximum gas pressure inside the cylinder is limited to 4 MPa. The cylinder head is made of cast iron and the allowable tensile stress is 40 N/mm². Determine the thickness of the cylinder head. The studs which are made of steel have allowable stress as 50 N/mm². Calculate i) Number of studs, ii) Nominal diameter, and iii) Pitch of studs. $K = 0.162$ [10]

OR

- Q8)** The following data is given for a four- stroke diesel engine : [16]

Cylinder bore = 250 mm, Length of stroke = 300 mm, speed = 600 rpm
 Indicated mean effective pressure = 0.6 MPa., Mechanical efficiency = 80 %
 Maximum gas pressure = 4 MPa
 Fuel consumption = 0.25 kg per BP per hr.
 Higher calorific value of fuel = 44000 KJ/kg

Assume that 5% of total heat developed in the cylinder is transmitted by the piston. The piston is made of gray C.I FG 200 ($S_{ut} = 200 \text{ N/mm}^2$ and $K = 46.6 \text{ W/m}^2\text{°C}$) and the factor of safety is 5. The temperature difference between the center and the edge of the piston head is 220°C.

- i) Determine the thickness of piston head by strength consideration and thermal consideration.
- ii) State whether the ribs are required, If so calculate the number and thickness of ribs.
- iii) State whether a cup is required in the top of piston head, If so calculate the radius of the cup.

Q9) a) Write a short note on design for manufacturing and assembly. [6]

- b) A tensile bar of length 300 mm is subjected to constant tensile force of 4 KN. If the factor of safety is 1.5, design the bar diameter, using Johnson's method with the objective of minimizing material weight using optimum material from the list given below. [10]

Material	Density(ρ) kg/m ³	Cost(c)Rs/Kg	Syt N/mm ²
Steel	7800	34	400
Aluminum alloy	2800	160	150
Titanium alloy	4500	2400	800

OR

Q10)a) A tensile bar of length 400 mm is subjected to the tensile force of 4.5 KN. Assume Factor of Safety 2. Design the bar with the objective of minimizing the material cost, out of the following materials: [10]

Material	Mass Density ρ kg/m ³	Material cost per Unit Mass c, Rs./kg	Yield Strength S_{yt} N/mm ²
Plain Carbon Steel	7500	16	130
Aluminum Alloy	3000	32	50
Titanium Alloy	4800	480	90
Magnesium Alloy	2100	32	20

- b) Explain design consideration in design for forgings. [6]

